Problem Set 2 杜乘旋 12209095

Q1

1. The presence of Omitted variable bias would depend on whether there are other factors that affect both land rental behavior and agricultural yield.
If there are omitted variables (eg. soil equality) that are correlated with both land_rental and yield, the estimated of β_1 and β_2 could be biased

2. $\hat{\beta_1} \xrightarrow{p} \beta_1 + \rho_{Xu}\frac{\sigma_u}{\sigma_x}$, $\rho_{Xu} = corr(X_i, u_i)$

If $\rho > 0$, $\hat{\beta_1} > \beta_1$

If $\rho < 0$, $\hat{\beta_1} < \beta_1$

eg. The omitted variable is the education attainment of householder.

```
. gen Edu = huzhu_edu
(542 missing values generated)

. regress yield rental_in_share rental_out_share Edu
```

Source	SS	df	MS	Number of obs	=	13,629
				F(3, 13625)	=	7.86
Model	3767747.24	3	1255915.75	Prob > F	=	0.0000
Residual	2.1768e+09	13,625	159765.492	R-squared	=	0.0017
				Adj R-squared	=	0.0015
Total	2.1806e+09	13,628	160006.793	Root MSE	=	399.71

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rental_in_share	.0520157	.0648314	0.80	0.422	-.0750628	.1790942
rental_out_share	.2316851	.0488358	4.74	0.000	.1359601	.3274101
Edu	.9399293	1.360007	0.69	0.490	-1.725872	3.605731
_cons	426.3138	9.82775	43.38	0.000	407.05	445.5775

```
. corr rental_in_share Edu
(obs=13,629)
```

	r~in_s~e	Edu
rental_in_~e	1.0000	
Edu	0.0105	1.0000

$\therefore \rho > 0$, $\hat{\beta_1} > \beta_1$

3. ① try to add the variables that may effect yield as much as possible

```
. regress yield rent_in rent_out rental_in_share rental_out_share household_operating_costs Edu fertiliser_price diesel_pric
> e firm_price agrochemicals household_income interest_income household_expenditures gender age health_condition
```

Source	SS	df	MS	Number of obs	=	267
				F(16, 250)	=	1.41
Model	431203.079	16	26950.1925	Prob > F	=	0.1388
Residual	4791406.77	250	19165.6271	R-squared	=	0.0826
				Adj R-squared	=	0.0238
Total	5222609.84	266	19633.8716	Root MSE	=	138.44

Variables

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rent_in	-2.504231	2.907923	-0.86	0.390	-8.23138	3.222919
rent_out	28.64993	21.71163	1.32	0.188	-14.11108	71.41095
rental_in_share	.1512069	.1391197	1.09	0.278	-.1227892	.425203
rental_out_share	-1.639185	1.576797	-1.04	0.300	-4.744685	1.466315
household_operating_costs	.0000806	.0006742	0.12	0.905	-.0012472	.0014084
Edu	-2.327571	3.736017	-0.62	0.534	-9.685651	5.030509
fertiliser_price	-9.991382	13.5036	-0.74	0.460	-36.5867	16.60393
diesel_price	.5624932	1.075682	0.52	0.601	-1.556061	2.681047
firm_price	4.302249	2.168969	1.98	0.048	.0304676	8.57403
agrochemicals	.2741959	.2119785	1.29	0.197	-.1432954	.6916873
household_income	.0004268	.0002924	1.46	0.146	-.0001491	.0010028
interest_income	.0007929	.0013771	0.58	0.565	-.0019194	.0035052
household_expenditures	-.000368	.0002401	-1.53	0.127	-.0008408	.0001049
gender	-3.033955	58.13414	-0.05	0.958	-117.5291	111.4611
age	.2002127	.9623441	0.21	0.835	-1.695123	2.095548
health_condition	1.170446	.4917961	2.38	0.018	.2018545	2.139038
_cons	269.3923	84.73517	3.18	0.002	102.5065	436.2781

4. If rent-in land increases by 1% percentage point, yield will decrease 0.2504

5.

```
. test Edu

 ( 1)  Edu = 0

       F(  1,    252) =    0.22
            Prob > F =    0.6416
```

$H_0: Edu = 0 \quad H_1: Edu \neq 0$

$\because 64.16\% > 5\%$

$\therefore$ Can't reject H_0

$\therefore$ can't approve Economist A's opinion.

6.

```
. test rental_in_share = rental_out_share

 ( 1)  rental_in_share - rental_out_share = 0

       F(  1,    252) =    0.17
            Prob > F =    0.6829
```

$\because 68.29\% > 10\%$

$\therefore$ Can't reject H_0

$\therefore$ can't reject B's opinion that renting in and renting out have the same effects on agricultural productivity.

7. in Question(3), $R^2 = 0.086$; in original regression, $R^2 = 0.0017$

$\therefore$ higher than original regression.

$\because$ in (3), has more variables than original.

```
. regress yield rental_in_share rental_out_share

      Source |       SS           df       MS      Number of obs   =    14,171
-------------+----------------------------------   F(2, 14168)     =     12.37
       Model |  3856365.5         2  1928182.75   Prob > F        =    0.0000
    Residual |  2.2092e+09   14,168  155931.491   R-squared       =    0.0017
-------------+----------------------------------   Adj R-squared   =    0.0016
       Total |  2.2131e+09   14,170  156181.633   Root MSE        =    394.88

------------------------------------------------------------------------------
       yield | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
rental_in_s~e |   .0533957   .0639909     0.83   0.404    -.0720348    .1788262
rental_out_~e |   .2349335   .0479369     4.90   0.000     .140971     .3288961
       _cons |   431.3688   3.35498     128.58   0.000     424.7926     437.945
------------------------------------------------------------------------------
```

R^2 is a statistical measure used to access the goodness of fit of a regression model, indicating the proportion of the variability in the dependent variable that is explained by the model.

However, R^2 cannot detect omitted variable bias and doesn't consider Model's complexity.

Q2

1. I think $E[u_i|X_i] \neq 0$. Because the treatment is not random, it is based on the village's location (coastal or island), and there may be unobserved factors related to the village's location that affect yields.
   $\therefore \beta_1$ may also be biased.

2. The unbiasedness of the OLS estimator depends on whether the error term, u_i, is correlated with X_i or W_i.
   a. $\because X_i$ denote the binary variable
      $\therefore$ is not systematically related to the error term
      $\therefore \beta_1$ remains unbiased. and $E(u_i|X_i, W_i)$ does not depends on X_i.

   b. $E(u_i|X_i, W_i)$ does not depends on W_i.
   $\because W_i$ denote a binary variable indicates whether the village in the coastal region or not.
   Because the omitted variables could affect Y_i in u don't correlated with W_i.

Q3

1. $\min \sum_{i=1}^{n} \hat{u}_i^2 = \min \sum_{i=1}^{n} (Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})^2$

2. OLS first order conditions: $\frac{\partial y}{\partial \beta_1} = -2\sum_{i=1}^{n} X_{1i}(y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$
   $\frac{\partial y}{\partial \beta_2} = -2\sum_{i=1}^{n} X_{2i}(y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$

3. Suppose $\sum_{i=1}^{n} X_{1i}X_{2i} = 0$, $\sum_{i=1}^{n} X_{1i}(Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$
   $\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_{2i}Y_i - \hat{\beta}_1 \sum_{i=1}^{n} X_{1i}X_{2i}}{\sum_{i=1}^{n} X_{1i}^2} = \frac{\sum_{i=1}^{n} X_{1i}Y_i}{\sum_{i=1}^{n} X_{1i}^2}$

4. If $\sum_{i=1}^{n} X_{1i}X_{2i} \neq 0$, $\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_{2i}\sum_{i=1}^{n} X_{1i}Y_i - \sum_{i=1}^{n} X_{1i}\sum_{i=1}^{n} X_{1i}Y_i}{\sum_{i=1}^{n} X_{1i}^2 \sum_{i=1}^{n} X_{2i}^2 - (\sum_{i=1}^{n} X_{1i}X_{2i})^2}$

5. $\because \frac{\partial \beta}{\partial \beta_0} = -2\sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_{1i} - \beta_2 X_{2i}) = 0$
   $\hat{\beta}_0 = \frac{1}{n}\sum_{i=1}^{n} (Y_i - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})$
   $\therefore \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2$

   b. $\sum_{i=1}^{n} (X_{1i} - \bar{X}_1)[X_{2i} - \bar{X}_2) = 0$
   show $\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_{1i} - \bar{X}_1)^2}$
   $\because Cov(X_{1i}, X_{2i}) = 0$ $\therefore \hat{\beta}_1 = \frac{\sum_{i=1}^{n} (X_{1i} - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (X_{1i} - \bar{X})^2}$

   $\therefore$ is the same as β_1 from the regression that omits X_2.